PARASITIC MODULES HAVE VANISHING COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is written for the fppf topology and states explicitly that the other cases are proved in the exact same way; the delivered text keeps that organisation and no case was expanded or added.

Editorial note. The counted spans carry 2 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Parasitic presheaves of modules. Same chapter, Definition definition-parasitic, invoked in the statement at lines 4010-4016. Fixes the hypothesis on the presheaf in both parts.
- The Cech vanishing criterion for a subset of a site with a collection of coverings. Cohomology on Sites, Lemma sites-cohomology-lemma-cech-vanish-collection, invoked at proof lines 4024-4028. Turns the vanishing of the presheaf on a suitable subset of the site into the vanishing of all cohomology.
- The definitions of the five topologies in question and of their coverings. Standard, used to identify the objects on which the hypothesis is assumed at proof lines 4022-4024. Determines which objects of the site form the subset to which the vanishing criterion is applied.

Source reference. Original lemma 03F9: sources/stacks-chow/tags.txt, source lines 4484–4484. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $\tau \in \{Zar, étale, smooth, syntomic, fppf\}$. Let $\mathcal{G}$ be a presheaf of $\mathcal{O}$-modules on $(Sch/S)_\tau$.*

(1) *If $\mathcal{G}$ is parasitic for the τ-topology, then $H^p_\tau(U, \mathcal{G}) = 0$ for every U open in S, resp. étale over S, resp. smooth over S, resp. syntomic over S, resp. flat and locally of finite presentation over S.*

(2) *If $\mathcal{G}$ is parasitic then $H^p_\tau(U, \mathcal{G}) = 0$ for every U flat over S.*

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Proof. Proof in case $\tau = fppf$; the other cases are proved in the exact same way. The assumption means that $\mathcal{G}(U) = 0$ for any $U \to S$ flat and locally of finite presentation. Apply Cohomology on Sites, Lemma 03F9 to the subset $\mathcal{B} \subset \mathrm{Ob}((Sch/S)_{fppf})$ consisting of $U \to S$ flat and locally of finite presentation and the collection Cov of all fppf coverings of elements of $\mathcal{B}$. □